Requirements Document: Autonomous Magnetic Chessboard

ECE445L Fall 2026 Team: Member A, Member B, Member C, Member D

Lab 7A, preliminary version, October 15, 2026

1 Overview

1.1 Objectives: Why are we doing this project?

Playing chess against a computer usually means a screen. Our goal is a physical chessboard that plays back: a single player moves real pieces, and the board senses the move, computes a reply, and slides its own piece into place with a hidden magnet. The end user is a chess player who wants a physical opponent, or a spectator who wants to watch the board play itself. The educational objectives are stepper motor control, timer and GPIO interrupts, SPI device drivers, sensor arrays, power design for inductive loads, and fitting a game engine into a small microcontroller.

1.2 Process: How will the project be developed?

Each subsystem is built and unit tested separately on its own LaunchPad and breadboard in Lab 7A, laid out on one PCB with power and enclosure in Lab 7B, and integrated with the high-level game software in Lab 10. Code lives in the team GitHub repository; each subsystem is developed on its own branch and merged by pull request after another member reviews it. The team meets twice a week and communicates in a shared group chat that we will show the TA.

1.3 Roles and Responsibilities: Who will do what? Who are the clients?

Owner	Responsibility
Member A	Gantry: stepper drivers, homing limit switches, timer-driven step generation, path planning.
Member B	Magnet carriage and mechanics: electromagnet drive circuit, board stack-up, pieces, enclosure.
Member C	Board sensing: 64 Hall sensors, shift-register chain, move detection.
Member D	User interface and game: ST7735 display, buttons, move legality, chess engine.

Everyone shares schematic and PCB layout work in Lab 7B. The clients are the ECE445L TA and professors, and the students who judge the Lab 10 design competition.

1.4 Interactions with Existing Systems

The system uses one MSPM0G3507 (M0G3507QDGS32RQ1 on the final PCB, the LP-MSPM0G3507 LaunchPad during development) as its only processor, programmed with TI DriverLib and the class

starter files. The ST7735 driver is reused from earlier labs. Parts footprints come from the ECE445L KiCad library where possible. The board runs from a single 12 V wall adapter. There is no network connection.

1.5 Terminology

Gantry: XY mechanism under the board that carries the magnet. **Homing**: driving each axis to a limit switch at power-up to find position zero. **Microstepping**: driver mode that splits a full motor step into smaller steps for smoother motion. **Hall effect sensor**: chip whose output changes in a magnetic field; *omnipolar* means either magnet pole triggers it. **Shift register**: chip that loads 8 parallel inputs and clocks them out serially. **Ply**: one move by one side; engine search depth is counted in plies.

1.6 Security

The device stores no user data and has no wireless or network interface, so the attack surface is limited to physical access to the debug port.

2 Function Description

2.1 Functionality: What will the system do precisely?

1. At power-up the gantry homes both axes and the board scans all 64 squares. If the pieces are not in the starting position the display lists the squares that are wrong.
2. The player chooses a color and difficulty with the buttons.
3. The player moves a piece by hand. The Hall sensor grid detects which squares changed; when the board is stable the move is decoded and checked for legality. Illegal moves are shown on the display.
4. The engine computes a reply and the display shows it. The gantry drives the magnet under the piece, energizes the electromagnet, and slides the piece along the lines between squares so it does not hit other pieces. Captured pieces are moved to storage squares beside the board.
5. After the move, the board rescans to confirm the piece arrived; if it did not, the display asks the player to fix it.
6. Special moves (castling, en passant, promotion) and end of game (checkmate, stalemate) are detected and shown on the display.

The system has at least three inputs (Hall sensor grid, buttons, limit switches), three outputs (two stepper motors, electromagnet, display), and two interrupt service routines (timer interrupt generating step pulses; GPIO interrupt for buttons and limit switches).

2.2 Scope: List the phases and what will be delivered in each phase

Phase	Delivered
Lab 7A	Requirements, schematic, BOM, pin allocation; every subsystem working on a breadboard with a unit test project.
Lab 7B	PCB layout, power supply design, enclosure plan.
Lab 10	Integrated board playing a full game against a person; competition demo.
Stretch	Automatic reset of pieces to the start position; a small quantized neural-network evaluation and a self-play mode where two engines play each other. Not required for the core project.

2.3 Prototypes: How will intermediate progress be demonstrated?

Lab 7A: one LaunchPad per subsystem. (1) Gantry moves the carriage to any square and returns to home. (2) Electromagnet pulls the heaviest piece through the real board top. (3) Hall grid occupancy map shown on the display while pieces are placed and lifted. (4) Display and buttons run the menu and the engine answers a scripted list of moves. Lab 10: the integrated board plays a full game.

2.4 Performance: Define the measures and how they will be determined

Measure	Target
Carriage position error at a square center after 50 moves without re-homing (calipers)	$\leq 2\,\text{mm}$
Time for the board to make a typical one-square to four-square move (stopwatch)	$\leq 8\,\text{s}$
Neighboring pieces disturbed while moving a piece (100 test moves)	0
Time to scan all 64 squares (scope on the latch pin)	$\leq 1\,\text{ms}$
Occupancy detection errors over 200 piece placements	0
Engine reply time on the MSPM0 at 80 MHz (timer measurement)	$\leq 5\,\text{s}$
Illegal player moves detected (scripted test set)	100%
Peak current from the 12 V supply (bench supply meter)	$\leq 3\,\text{A}$

2.5 Usability: Describe the interfaces

Inputs: the player moves pieces by hand; three buttons (select, confirm, new game) navigate the menus. Outputs: the 1.8-inch color display shows prompts, the last move, the engine's move, and errors; the board moves its own pieces. Power: one 12 V adapter and a power switch.

2.6 Safety: Explain any safety requirements and how they will be measured

The 12 V electromagnet is switched by a MOSFET with a flyback diode, and firmware limits how long it can stay on so the coil does not overheat (coil temperature measured after a 10-minute game). The gantry is fully enclosed under the board so fingers cannot reach the belts. Small neodymium magnets are sealed inside the pieces. A label warns users with pacemakers to keep clear of the magnets.

3 Deliverables

3.1 Reports

Lab 7A, Lab 7B and Lab 10 reports, including unit test results and the current-consumption estimate.

3.2 Audits

TA checkouts for Labs 7A, 7B and 10; preliminary PCB review by the TA and professor before ordering.

3.3 Outcomes

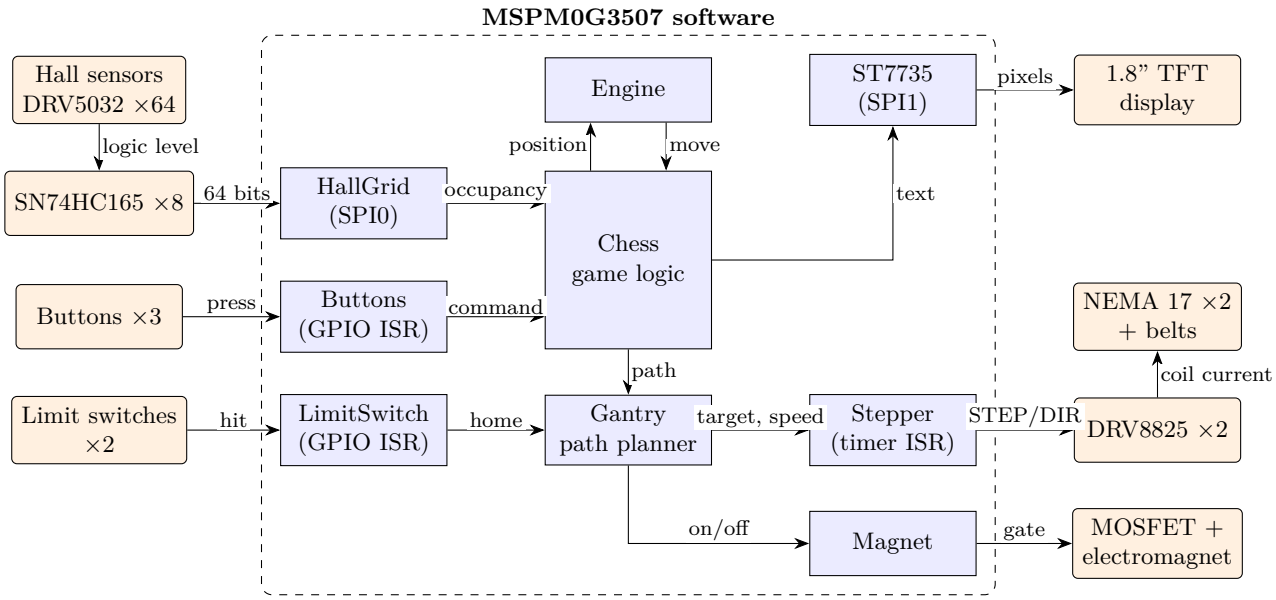
A working autonomous chessboard, its KiCad schematic and PCB, a BOM (current estimate $151.35 total, $79.00 counting toward the competition limit, before power and enclosure), and all source code in GitHub.

Open questions for the TA

- The 64-sensor grid (about 13 in square) is larger than the 30 sq in PCB. Can the sensors sit on separate carrier boards, with the shift registers on the main PCB?
- The handout gives both $60 and $80 as the competition parts limit. Which applies?

4 Data Flow Graph

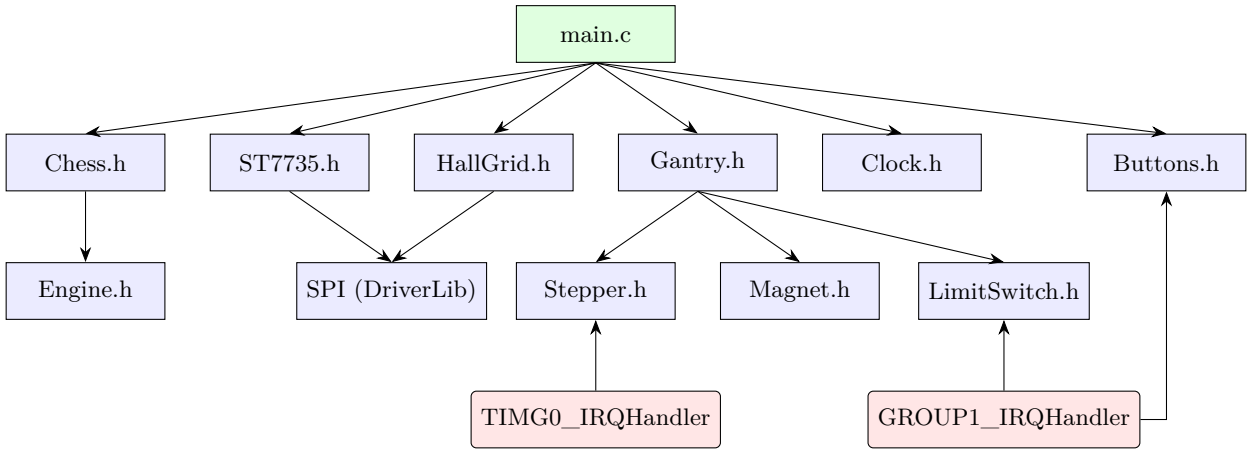
Orange boxes are hardware, blue boxes are software modules on the MSPM0G3507 (inside the dashed line). Arrows show the data passed between them.



The motors carry the electromagnet under the board; its field drags the selected piece, and the Hall sensors see the piece arrive, closing the loop.

5 Call Graph

Module names match the header files. Arrows point from caller to callee; red boxes are interrupt service routines.



`TIMG0_IRQHandler` emits one STEP pulse per interrupt at the rate set by `Gantry`; `GROUP1_IRQHandler` services button presses and limit-switch hits.

6 Preliminary Pin Budget

Device	Signals	Pins
Hall grid (74HC165 chain)	SPI0 SCLK, SPI0 POCI, SH/LD latch	3
Stepper drivers	STEP ×2, DIR ×2, shared ENABLE	5
Electromagnet	PWM-capable gate output	1
Limit switches	2 GPIO inputs with interrupts	2
ST7735 display	SPI1 SCLK, SPI1 PICO, CS, D/C, RESET	5
Buttons	3 GPIO inputs with interrupts	3
Debug UART (optional)	TX, RX	2
Total	of 26 Port A pins available on the M0G3507QDGS32RQ1	**21**

Exact pin numbers for the LaunchPad and the VSSOP-32 package go in `MSPM0_Pins_Assignments.xlsx`.